Muhammad Azeem Bhatti

AI/ML Engineer | Medical Imaging & Speech AI

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Portfolio: ai-collective-lab.github.io/portfolio-muhammad-azeem-bhatti

AI/ML engineer building production speech, vision, and LLM systems, from Urdu-English transcription and meter-reading OCR to news-intelligence pipelines, with a research focus on medical-imaging ML.

EDUCATION

MS, Artificial Intelligence — Air University, E-9 Campus, Islamabad | Sep 2025 - Aug 2027 (Expected)
CGPA: 3.97 / 4.0

BS, Electrical Engineering (Electronics) — Air University, E-9 Campus, Islamabad | Sep 2015 - Jun 2019

EXPERIENCE

AI Intern — Golden Gate Innovations | Jul 2026 - Present

Speech Transcription & Translation Pipeline (Urdu-English) — Solo

- Built a production Urdu-English speech transcription and translation pipeline (faster-whisper large-v3 + Fireworks GLM-5.2), cutting translation WER from ~62% to ~46% with a translate-critique-refine loop; reached 28.9% WER on a human-annotated benchmark.
- Added real-time speaker diarization (pyannote.audio) to the GPU inference service, resolving CUDA and dependency conflicts in production.
- Built a bilingual PDF transcript export with right-to-left text shaping; deployed the GPU service on RunPod with Docker and zero-downtime rollouts.

OSINT News Intelligence Pipeline (Pakistan-Interest News Monitoring System) — Solo

- Built an end-to-end Python OSINT pipeline that collects, deduplicates, and classifies news from 8+ live sources (RSS, Reddit, government portals) into SQLite.
- Added a Groq LLM classifier for genre and national-interest tagging with confidence scores on 500+ articles.
- Designed pluggable source connectors with content-hash deduplication and APScheduler batch runs.

K-Electric Automated Meter Reading — Solo

- Built an automated meter-reading system that localizes K-Electric utility meters in field photos with YOLO11n rotated bounding boxes (OBB) and reads them with a ResNet18 + BiLSTM (CRNN) OCR model.
- Added a five-model voting ensemble: 98.2% exact readings on 606 held-out crops.
- Added auto-accept stage: 95.7% of crops at 99.5% accuracy, reducing manual review.

PROJECTS

LatentGAT

Open-Source · Sep 2026

Built a graph-attention 3D U-Net for brain tumor segmentation: Dice 0.869 on BraTS2021 and two unseen cohorts (BraTS-Africa; UCSF-PDGM, n=239).
Code: github.com/AI-Collective-Lab/latentgat-training-dashboard

Effectiveness of Balancing Plant's Dataset using GAN

Balanced a 38-class crop disease dataset with DCGAN samples: 98.19% accuracy, 0.982 macro-F1.
Code: github.com/AI-Collective-Lab/Effectiveness-Of-Balancing-Dataset-On-Classification

SKILLS

ML/DL & Research: Python, PyTorch, scikit-learn, GANs, U-Net, Graph Attention Networks, YOLO, CRNN OCR

NLP & Speech AI: LLM APIs, faster-whisper, pyannote.audio (Speaker Diarization), Machine Translation

Deployment & Tools: Docker, RunPod (GPU inference), SQLite, SQLAlchemy, Streamlit, APScheduler, Git, GitHub